

Multi-Modal Structured Table Parsing for RAG Systems

In industrial-grade LLM applications, multi-modal structured table parsing has emerged as a critical component in Retrieval-Augmented Generation (RAG) systems. Traditional text-only parsing methods often fail to accurately identify row-column structures in PDF tables, resulting in severe semantic distortion. This study proposes a two-dimensional matrix extraction algorithm using pdfplumber, which automatically converts PDF tables into standard Markdown format, significantly improving retrieval accuracy in RAG pipelines.

Model Name	Exp Group	Ctrl Group
Baseline	82.5	12.4
Advanced RAG	98.9	11.2

Experimental results demonstrate that the Advanced RAG model achieves a significantly higher experimental group score (98.9) compared to the Baseline model (82.5), validating the effectiveness of structured table parsing in enhancing RAG performance. The control group score (11.2) further confirms that structured parsing substantially reduces retrieval noise. These findings provide important theoretical foundations and practical references for data governance engineering in production-grade RAG systems.